

Jovana Kondic

✉ jkondic@mit.edu | 🔗 linkedin.com/in/jovanakondic | 🎓 Jovana Kondic

Education

Massachusetts Institute of Technology

PhD in Electrical Engineering and Computer Science

Cambridge, MA

Current - May 2027

- Advised by Prof. Aude Oliva

Massachusetts Institute of Technology

MS in Electrical Engineering and Computer Science

Cambridge, MA

Sept 2021 - January 2024

- Thesis topic: Monte Carlo Methods for Motion Planning and Goal Inference
- Coursework: algorithms for inference, deep reinforcement learning, robotic manipulation, smart control of decarbonized power systems

Princeton University

BSE in Electrical Engineering; certificates in Cognitive Science, Robotics, and ML & Statistics

Princeton, NJ

Sept 2017 - June 2021

- Thesis topic: Towards Socially Aware Robot Learning: Inferring Human Objectives and Latent Safety Preferences from Observations
- Coursework: safety-critical robotics, computational neuroscience, game theory, probability & stochastic systems, animal learning, embedded computing, biomechanics, moral philosophy

Research Experience

Computational Perception and Cognition Lab

Massachusetts Institute of Technology

Cambridge, MA

April 2024 - current

- Vision Language Models (VLMs) for chart understanding and synthetic data generation

MetaConscious Group

Massachusetts Institute of Technology

Cambridge, MA

March 2023 - December 2023

- Developed a brain-inspired memory architecture to facilitate long-horizon reasoning in LLM-powered agents in multi-agent environments

Algorithmic Alignment Group

Massachusetts Institute of Technology

Cambridge, MA

September 2021 - February 2023

- Implemented MC sampling algorithms for diversity-aware near-optimal motion planning and trajectory prediction

Safe Robotics Lab

Princeton University

Princeton, NJ

September 2020 - May 2021

- Proposed a framework for 1) multi-objective modeling of human behavior and 2) Bayesian inference of task objectives and risk tolerances in safety-critical human-robot interaction scenarios

Intelligent and Interactive Autonomous Systems Group

Stanford University

Palo Alto, CA

June 2020 - Oct 2020

- Achieved up to + 50% performance boost (compared to best baseline) in two-player coordination tasks, by helping develop a modular RL policy network that learns separate representations of task rules and partner-specific conventions

Work Experience

IBM Research/ MIT-IBM Watson AI Labs

Research Intern

Cambridge, MA

June 2025 - August 2025

- Co-led a chart understanding research project within the IBM Granite Vision Team, actively coordinating with 20+ researchers across 5 offices.

IBM Research/ MIT-IBM Watson AI Labs

Research Intern

Cambridge, MA

June 2024 - August 2024

- Designed and implemented a fully automated pipeline for large-scale, code-driven synthetic chart generation, enabling the creation of open-source chart understanding datasets that meet the scale requirements of foundation models (to be presented at ICCV 2025 workshop).

Massachusetts Institute of Technology

Teaching Assistant

Cambridge, MA

October 2023 - December 2023

- Collaboratively advised 14 student-led projects at the intersection of deep learning, Bayesian modeling, and neuroscience in course 9.58: *The Science of Intelligence* taught by Prof. Tomaso Poggio and Dr. Brian Cheung

Pillar VC

Venture Research Associate

Boston, MA

May 2023 - September 2023

- Identified and evaluated 8 potential investment opportunities within the applied ML sector through comprehensive market research and close collaboration with senior partners

Skills

	Languages	English, Italian, German, Serbian	* denotes limited experience
	Programming & Planning Languages	Python, Julia*	
	Libraries & Toolboxes	Numpy, PyTorch	
	Platforms & Frameworks	Github, HuggingFace, Langchain*	

Publications & Workshop Proceedings

- G. V. Team, L. Karlinsky, A. Arbelle, A. Daniels, A. Nassar, A. Alfassi, B. Wu, E. Schwartz, D. Joshi, **J. Kondic**, N. Shabtay, P. Li, R. Herzig, S. Abedin, S. Perek, S. Harary, U. Barzelay, A. R. Goldfarb, A. Oliva, B. Wiele, B. Bhattacharjee, B. Huang, C. Auer, D. Gutfreund, D. Beymer, D. Wood, H. Kuehne, J. Hansen, J. Shtok, K. Wong, L. A. Bathen, M. Mishra, M. Lysak, M. Dolfi, M. Yurochkin, N. Livathinos, N. Harel, O. Azulai, O. Naparstek, R. T. de Lima, R. Panda, S. Doveh, S. Gupta, S. Das, S. Zawad, Y. Kim, Z. He, A. Brooks, G. Goodhart, A. Govindjee, D. Leist, I. Ibrahim, A. Soffer, D. Cox, K. Soule, L. Lastras, N. Desai, S. Ofek-koifman, S. Raghavan, T. Syeda-Mahmood, P. Staar, T. Drory, and R. Feris (2025). **Granite Vision: a lightweight, open-source multimodal model for enterprise Intelligence**. arXiv: 2502.09927 [cs.CV]. [Link to paper](#).
- J. Kondic**, P. Li, D. Joshi, Z. He, S. Abedin, J. Sun, B. Wiesel, E. Schwartz, A. Nassar, B. Wu, A. Arbelle, A. Oliva, D. Gutfreund, L. Karlinsky, and R. Feris (2025). **ChartGen: Scaling Chart Understanding Via Code-Guided Synthetic Chart Generation**. *ICCV Workshop on Graphic Design Understanding and Generation, ICCV Workshop on Curated Data for Efficient Learning*. [Link to paper](#).
- K. I. Zhao, M. Naim, **J. Kondic**, M. Ernesto Cortes, J. Ge, S. Luo, G. R. Yang, and A. Ahn (2024). **Lyfe Agents: Generative Agents for Low-Cost Real-Time Social Interactions**. *ArXiv*. [Link to paper](#).
- J. Ge, K. Zhao, M. Ernesto Cortes, **J. Kondic**, S. Luo, M. Naim, A. Ahn, and G. R. Yang (2024). **Enhancing Understanding in Generative Agents through Active Inquiring**. *NeurIPS Workshop on Intrinsically-Motivated and Open-Ended Learning*.
- S. Gao, A. Fang, Y. Huang, V. Giunchiglia, A. Noori, J. R. Schwarz, Y. Ektefaie, **J. Kondic**, and M. Zitnik (2024). **Empowering biomedical discovery with AI agents**. *Cell* 187.22, pp. 6125–6151. ISSN: 0092-8674. DOI: <https://doi.org/10.1016/j.cell.2024.09.022>. [Link to paper](#).
- T. Zhi-Xuan, **J. Kondic**, S. Slocum, J. B. Tenenbaum, V. K. Mansinghka, and D. Hadfield-Menell (2023). **Bayesian Inverse Motion Planning for Online Goal Inference in Continuous Domains**. *ICRA Workshop on Cognitive Modeling in Robot Learning for Adaptive Human-Robot Interactions*. [Link to paper](#).
- A. Shih, A. Sawhney, **J. Kondic**, S. Ermon, and D. Sadigh (2021). **On the Critical Role of Conventions in Adaptive Human-AI Collaboration**. *ICLR*. [Link to paper](#).
- Y. Liu, A. G. Domel, S. A. Yousefsani, **J. Kondic**, G. Grant, M. Zeineh, and D. B. Camarillo (2020). **Validation and Comparison of Instrumented Mouthguards for Measuring Head Kinematics and Assessing Brain Deformation in Football Impacts**. *Annals of Biomedical Engineering* 48. [Link to paper](#).

Awards

2025	MIT-IBM Watson AI Lab Graduate Research Assistanship , supporting one year of fully funded doctoral research in AI	MIT
2021	Hewlett Packard Fellowship , merit-based award in the amount of \$90,154	MIT
2021	Sigma Xi Honors Society , inducted to the academic honors society for scientific research	Princeton
2020	Stanford Summer Undergraduate Research Fellowship , selected as one of 16 awardees from a pool of international applicants to participate in a fully-funded summer research program at Stanford	Stanford
2014	Google Science Fair , regional finalist	Google

Service & Leadership

2025	Co-organizer @ ICCV Workshop , co-hosted the First Workshop on Memory and Vision at ICCV 2025	MIT
2024	Mentor @ Undergraduate Research Opportunities Program , advising a student over the course of a semester on a project focused on evaluating large language model reasoning	MIT
2023	Executive Board member @ Graduate Application Assistance Program , led operations of departmental initiatives providing application guidance to students from underrepresented backgrounds	MIT
2020	Teaching Assistant @ Princeton EE/ Keller Center , collaborated with faculty to implement innovative teaching methods resulting in improved student retention and success	Princeton
2020	Peer Mentor @ Computer Science Council , guided first-year students through curriculum planning and academic support	Princeton
2019	Mentor @ Society of Women Engineers , engaged in panel discussions aimed at inspiring high school students to consider careers in STEM fields	Princeton